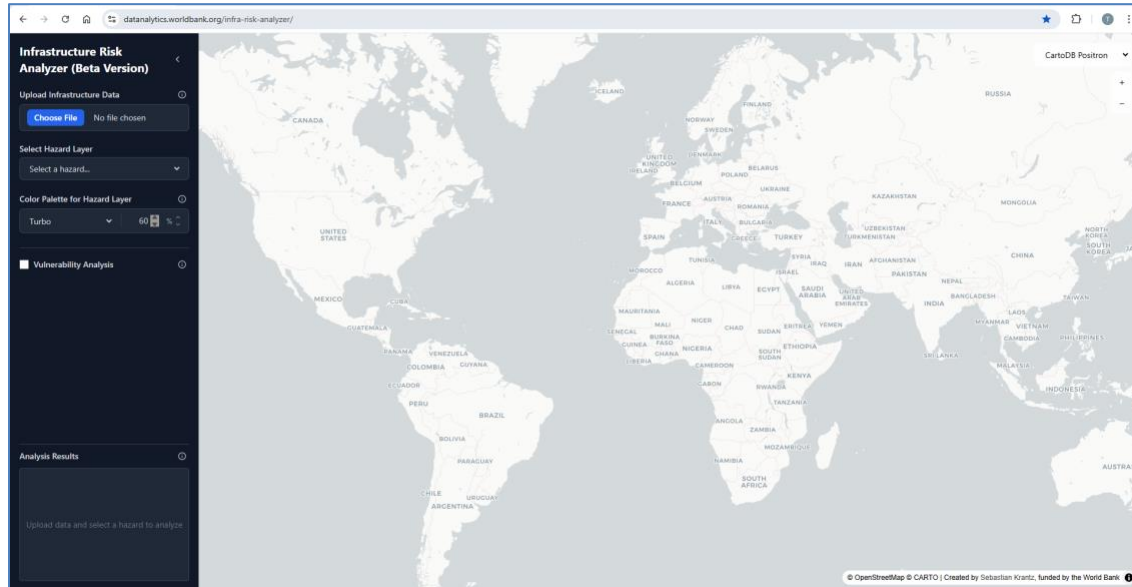


Infrastructure risk analyzer: Quick start guide

Beta Version, March 2026



About the tool

The [Infrastructure Risk Analyzer](#) is a secure web-based application developed by World Bank that automates climate and natural-hazard risk analysis for (digital) infrastructure assets. It is designed to help task teams and clients, infrastructure specialists, and consultants rapidly assess how exposed or vulnerable a portfolio of assets is to hazards such as flooding, high winds, or cyclones, without requiring specialist GIS skills.

The tool supports two types of analysis:

- **Exposure Analysis** identifies which infrastructure assets are located in areas where hazard intensity exceeds a threshold you define. This is a binary, spatial assessment: it tells you where assets are at risk, but does not estimate damage ratio or potential costs. It is a useful first-pass screening tool to flag high-risk assets early in project preparation.
- **Vulnerability and Damage Assessment** goes further by applying damage curves to estimate the economic cost of potential damage and losses. This approach produces asset-specific damage cost estimates that can inform investment prioritization and risk disclosure.

Who Is This Guide For?

This guide is written for World Bank task team members, infrastructure specialists, and consultants who want to conduct a rapid climate risk screening or more detailed risk analysis. You

do not need a GIS background to use the tool, but you should have access to basic location data for your infrastructure assets and a general understanding of the project context and planning horizon.

If you are unfamiliar with any of the concepts in this guide (such as damage curves or hazard scenarios), the explanatory notes within each section will help orient you. For complex or high-stakes assessments, consider consulting or engaging a climate risk specialist to review your inputs and interpret results.

Before You Begin: Data Checklist

Gather the following before opening the tool. Having these ready will make the process significantly faster.

What you need	Where to get it	Required for
Infrastructure asset data (location of assets as points or lines)	Your project team, national utility, or spatial planning agency	Exposure and vulnerability analysis
A clear sense of the project design life (e.g., 20 years, 50 years)	Project documentation	Selecting appropriate hazard scenarios
Damage curves for your asset type (CSV format)	See Section 5 for guidance	Vulnerability analysis only
Replacement cost estimates per unit of asset (e.g., per meter of cable, per tower)	Engineering cost estimates, World Bank unit cost databases, or regional benchmarks	Vulnerability analysis only

Understanding the Workflow

The tool follows a logical two-phase workflow. You can complete Phase 1 alone if you only need an exposure screening, or continue into Phase 2 for a full damage assessment.

- **Phase 1, Exposure Analysis:** Upload your assets → Select hazard and scenario → Set threshold → Review results
- **Phase 2, Vulnerability and Damage Assessment:** Enable vulnerability mode → Upload damage curve → Enter replacement costs → Review results

Both phases end with a map interface showing results, which can be explored in detail by maneuvering via zoom in/out function and by clicking on assets to gain specific information, as well as an export step.

Step-by-Step Guide

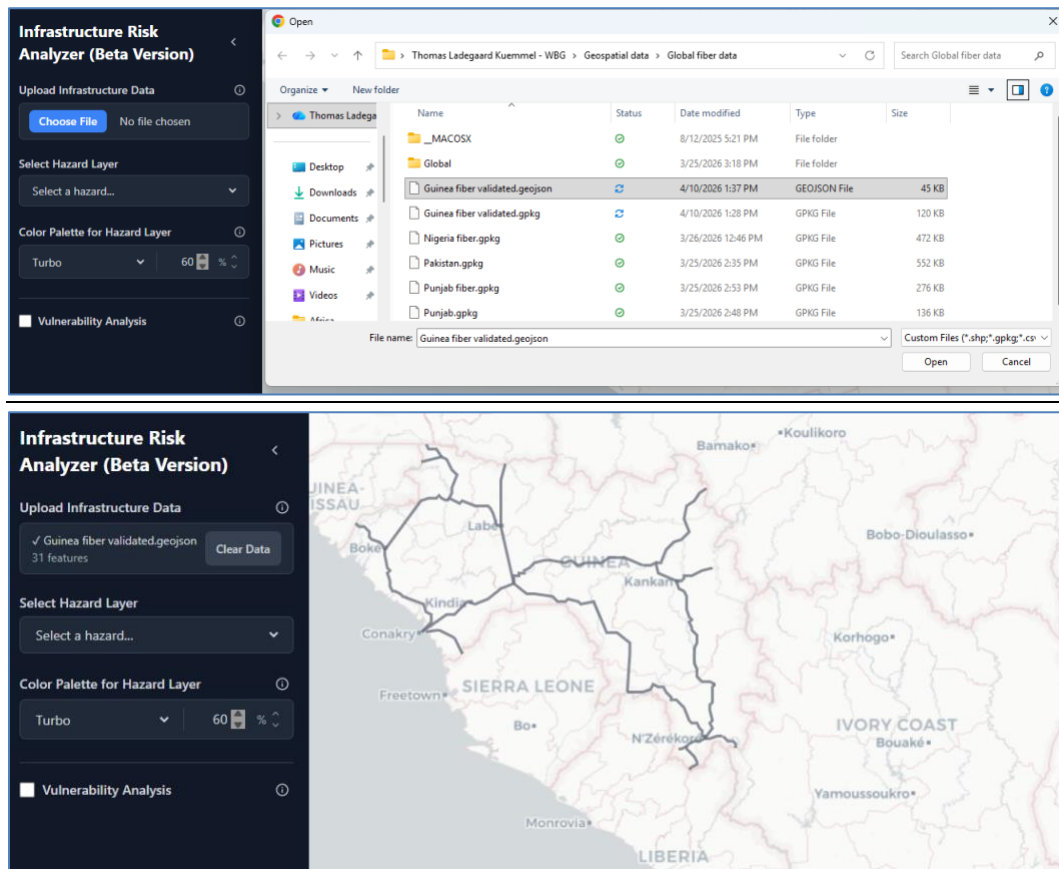
Step 1: Upload Your Infrastructure Data

Click 'Choose File' and select your infrastructure dataset. The maximum file size is 100 MB.

Once uploaded, your assets will appear as points or lines on the map. Check that they appear in the correct location before proceeding. The tool expects coordinates in the WGS84 system (EPSG:4326), which is the standard used by GPS and most online maps.

Accepted file formats:

Format	Notes
Shapefile (.shp)	Must be uploaded as a ZIP file containing all required components: .shp, .shx, .dbf, and optionally .prj
GeoPackage (.gpkg)	Single-file geospatial format. If your file contains multiple layers, the app will prompt you to select one
GeoJSON (.geojson)	Standard open format for geographic data
CSV (.csv)	Comma- or semicolon-delimited file with coordinate columns. Supported column name pairs: lat/lon, lat/lng, latitude/longitude, y/x (case-insensitive). CSV files are treated as point geometries only



Note: Please ensure the data is available in the correct format; otherwise, it will not display on the map, and an error message will be generated.

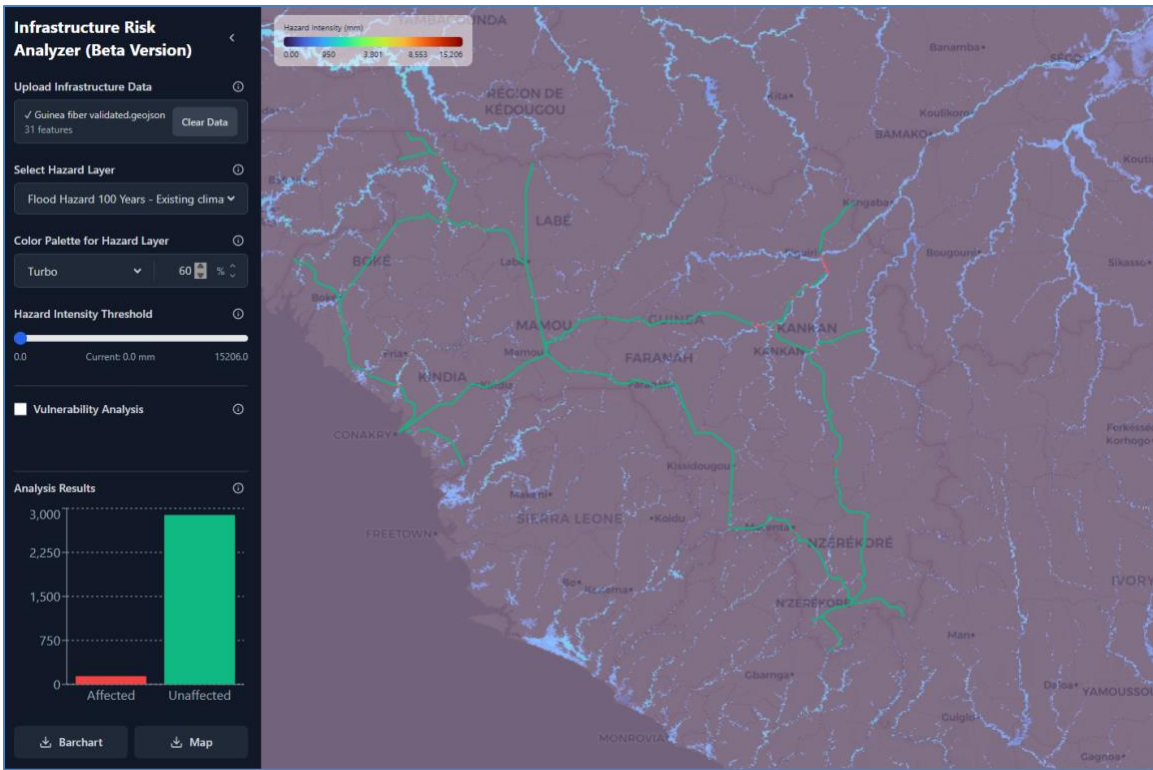
Step 2: Select Hazard Type and Scenario

Select the hazard type, return period and climate scenario relevant to your analysis from the dropdown menus. The tool offers a range of hazard layers (from <https://giri.unepgrid.ch/>) and scenarios representing different warming pathways and time periods.

Choosing the right scenario:

The scenario you select should reflect your project's planning horizon and risk appetite. As a general guide:

Project design life	Recommended scenarios to include
Up to 20 years	
20–50 years	
50 years or more	



Best practice: Always run the tool under at least two scenarios to understand the range of possible risk. A result that shows high exposure under the current baseline requires immediate attention, while exposure that only becomes significant under a high-warming scenario indicates a medium-term risk to monitor.

Hazard terminology

Types of hazards:

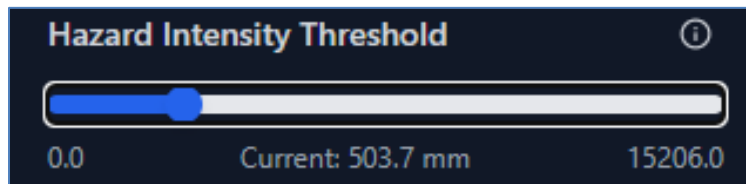
Return Period (Frequency): Represents the average time interval between events of a specific magnitude (e.g., a 100-year flood). A higher-severity event (e.g., a 100-year flood) is rarer (longer return period) and more severe than a lower-severity event (e.g., a 25-year flood). Return period estimation is based on the study of historical data of a specific event.

Probability: A 100-year return period means a chance of occurring in any given year, not that it occurs exactly every 100 years.

What the scenarios mean: **XX**

Step 3: Set Exposure Thresholds

Set the hazard intensity threshold above which an asset is considered exposed. For example, you might define a flood depth of 500 mm (50 cm) as the threshold at which a buried fiber optic cable or telecom tower is considered at risk of damage.



Why this matters: The threshold you choose determines what counts as "exposed." Too low a threshold will flag nearly all assets; too high will miss genuine risks. The right threshold depends on the engineering characteristics of your asset type.

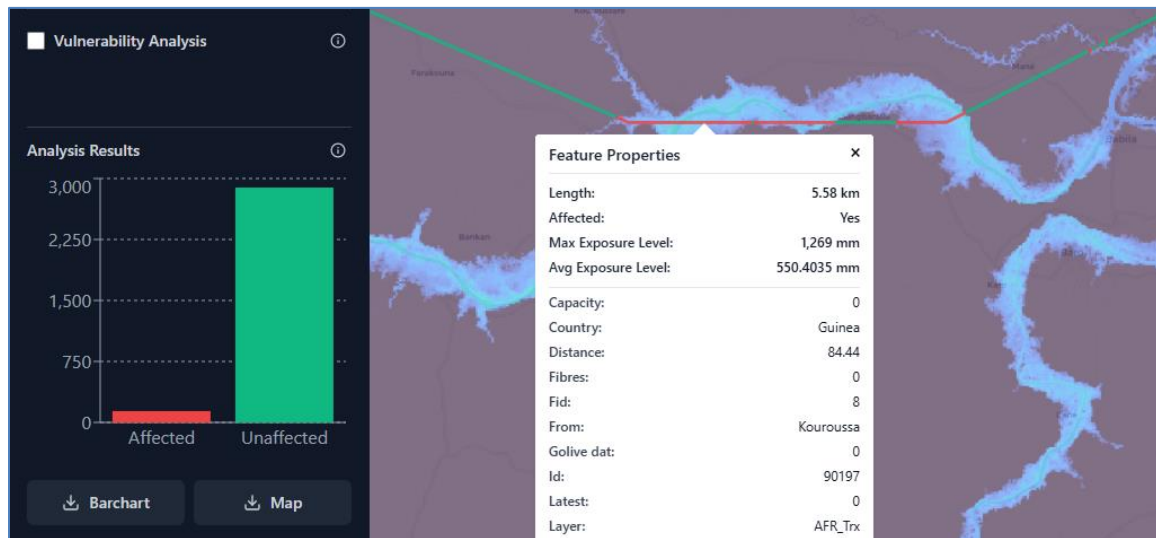
Suggested default thresholds by asset type (illustrative, adjust based on engineering specifications):

Asset type	Hazard	Suggested threshold	Rationale
Fiber optic cable (buried)	Flood depth	≥ 500 mm	Buried cables are generally tolerant of shallow flooding
Telecom tower	Flood depth	≥ 500 mm	Risk to ground-level equipment and foundations
Fiber optic cable (aerial)	Wind speed	≥ 20 m/s	Aerial lines at risk of structural failure above this range
Telecom tower	Wind speed	≥ 30 m/s	Standard design load thresholds for lattice towers

These are indicative starting points. If your project has specific engineering design standards, use those as the basis for threshold selection. When in doubt, run the analysis at multiple thresholds to understand sensitivity.

Step 4: Review Exposure Results

The tool will display bar charts showing the percentage of assets exposed to the selected hazard above your defined threshold (see step 3). You can click on individual points or line segments on the map to explore which specific assets are flagged as exposed.



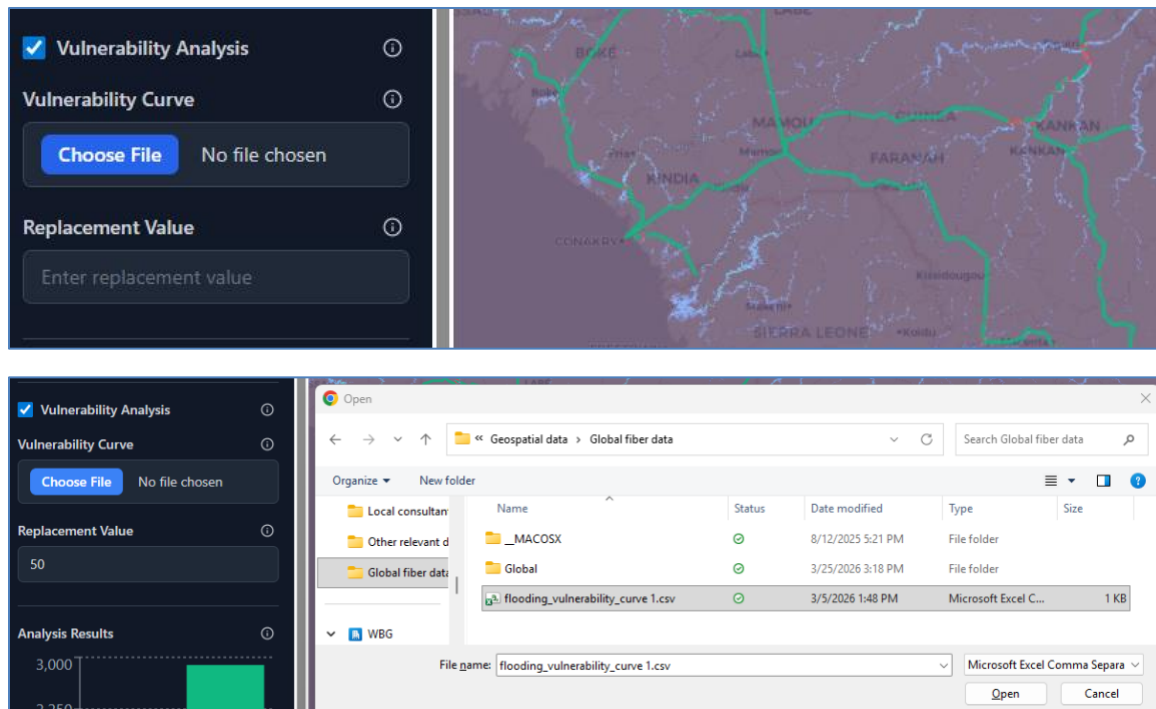
Interpreting your results:

- A significant exposure rate (e.g. >5 percent) under the existing climate suggests that assets face significant risk from present-day hazards, not just future climate change. This may require immediate engineering or siting responses.
- A significant exposure rate only under future high-warming scenarios suggests a medium- to long-term risk that should be tracked and factored into adaptive design or maintenance planning.
- A low exposure rate across all scenarios provides evidence that the asset portfolio is relatively well-located with respect to the hazard in question, though this should always be cross-checked against other hazard types.

Note: Exposure analysis is binary. An asset is either above or below the threshold, the tool does not differentiate between an asset that is just over the threshold and one that is deeply in the hazard zone. Use the vulnerability analysis in Phase 2 to capture the magnitude of risk.

Step 5: Enable Vulnerability Analysis and Upload a Damage Curve

To estimate potential economic losses, enable the vulnerability analysis module by checking the box in the tool interface. You will then be prompted to upload a damage curve.



What is a damage curve?

A damage curve (also called a fragility or vulnerability curve) describes the relationship between hazard intensity (e.g., flood depth in millimeters) and the proportion of an asset's value that is damaged (expressed as a ratio from 0 to 1). For example, a damage ratio of 0.3 at 500 mm flood depth means that an asset is expected to lose 30% of its replacement value when inundated to that depth.

File format: CSV with two columns, intensity and damage ratio.

Example:

```
intensity,damage_ratio
0,0.00
250,0.05
500,0.20
750,0.50
1000,0.80
1500,1.00
```

Where to obtain damage curves:

- Navigate to <https://github.com/SebKrantz/VulnerabilityCurves>, which contains curves from Nirandjan, S. et al: A spatially-explicit harmonized global dataset of critical infrastructure vulnerability curves, Nat. Hazards Earth Syst. Sci., 24, 4341–4369, 2024. DOI: 10.5194/nhess-24-4341-2024. Download the repo by clicking on Code -> Download ZIP.
- The World Bank's GFDRR and affiliated teams have developed standard damage curves for several infrastructure types. Contact your regional DRM or infrastructure specialist.

- The JRC (Joint Research Centre) Global Flood Depth-Damage Functions database provides open-access curves for a range of asset types and regions.
- If no standard curve exists for your asset type, a simple linear or piecewise function based on engineering judgment is a reasonable starting point, document the assumption clearly in any outputs you share.

If you do not have a damage curve and cannot obtain one, you can still use the exposure analysis from Phase 1 to support risk disclosure or investment prioritization.

Step 6: Enter Replacement Costs

Enter the unit replacement cost for your asset type. This figure is multiplied by the damage ratio from the curve to estimate economic losses per asset.

Format: Enter costs in USD per unit (per meter for linear assets such as cables or roads; per unit for discrete assets such as towers or substations).

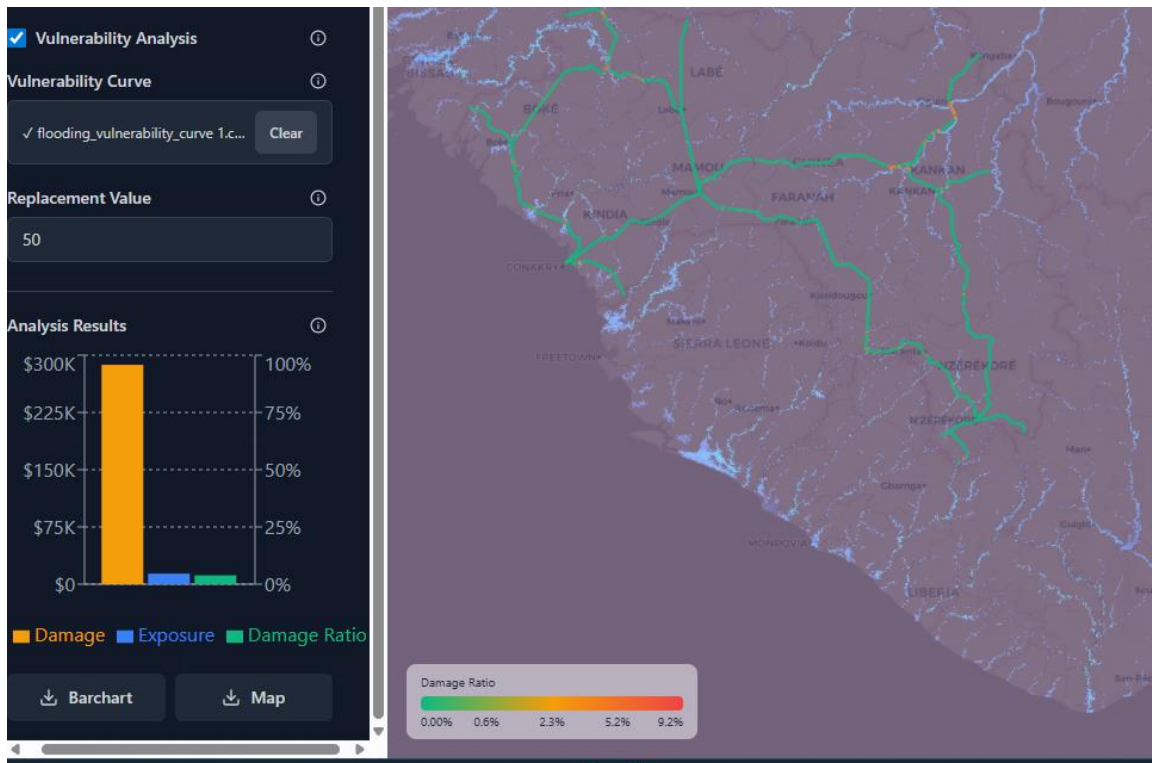
Reference cost ranges (illustrative, verify against project-specific estimates):

Asset type	Approximate unit cost
Fiber optic cable (buried)	\$20–50 per meter
Fiber optic cable (aerial)	\$10–25 per meter
Telecom tower (greenfield)	\$70,000–120,000 per tower

Important caveats: These figures are illustrative global averages. Actual costs vary significantly by country, terrain, contractor market, and specification. Where possible, use project-specific engineering cost estimates. Results are sensitive to cost assumptions – running the analysis with a low and high cost estimate will show you the range of potential losses.

Step 7: Review Vulnerability and Damage Results

The tool displays results as bar charts showing the percentage of asset value damaged and the estimated economic cost of damage under each scenario. You can click on individual assets on the map to see asset-level damage estimates.

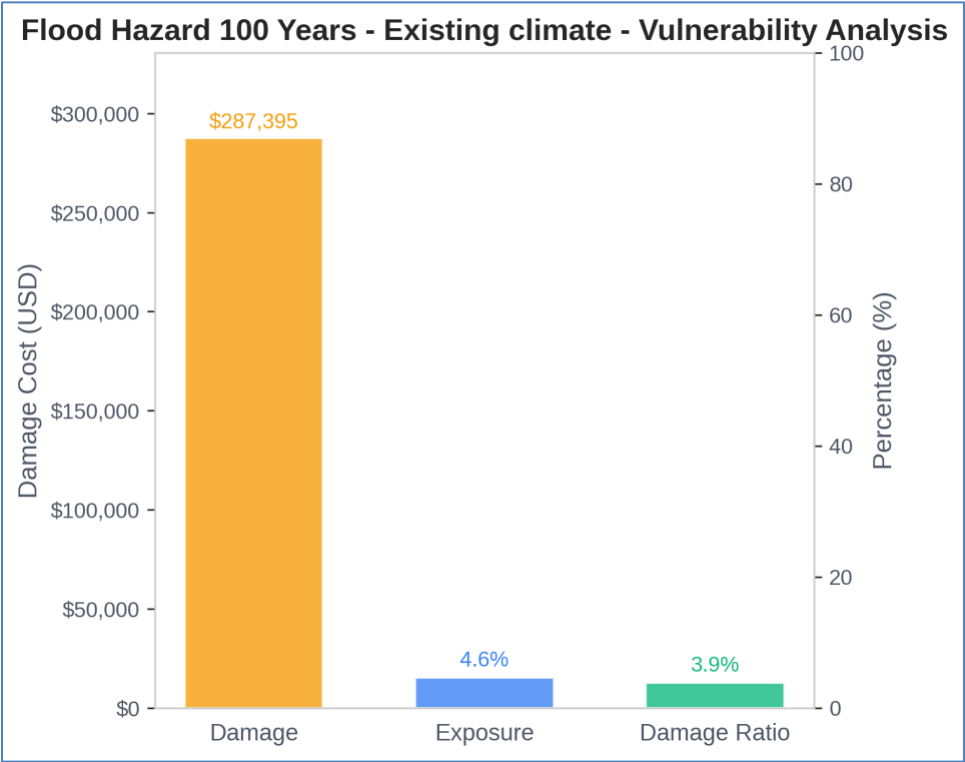
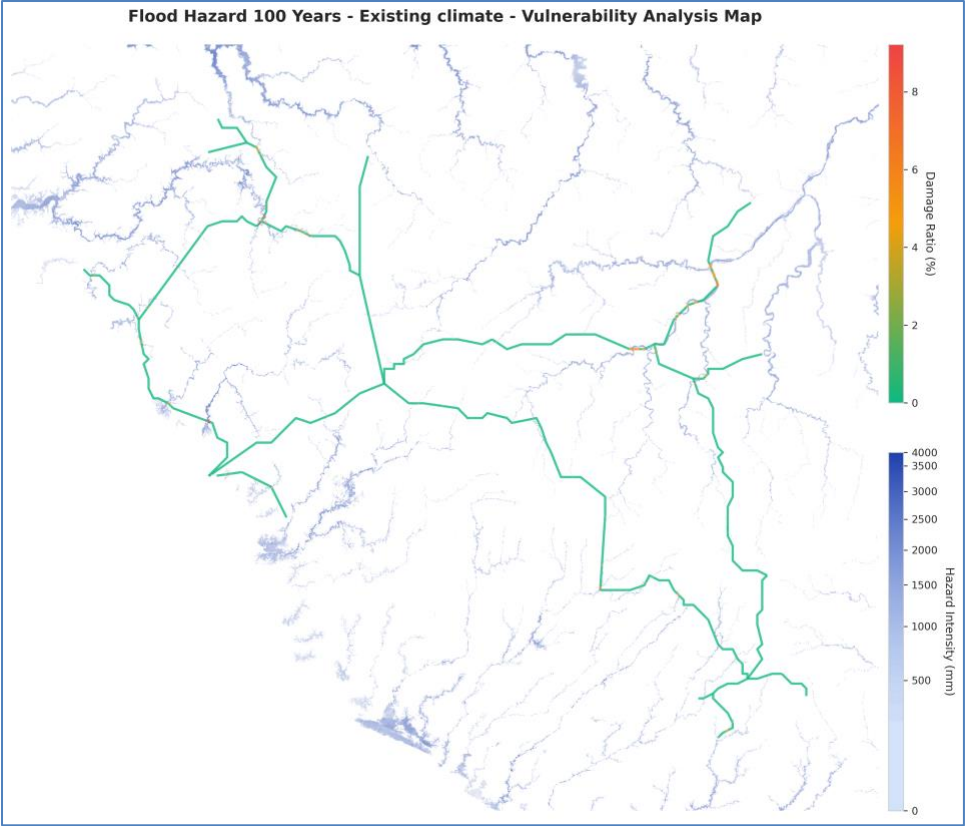


Interpreting your results:

- Review results across hazard scenarios to understand how damage costs change as climate conditions intensify. A large jump between the existing and future climate scenarios indicates that climate change is a major driver of future risk, rather than the asset simply being poorly located today.
- Pay attention to which individual assets drive the highest damage costs. These are priority candidates for design adaptation, relocation, or enhanced protection measures.
- Treat damage cost estimates as order-of-magnitude figures rather than precise valuations. They are derived from generalized damage curves and average cost estimates, and should be interpreted as indicative of relative risk across assets rather than as engineering-grade loss estimates.

Step 8: Export Maps and Results

Use the export function to download maps and bar charts. Outputs can be used directly in technical reports, climate risk screenings, or project presentations.



Tips for using outputs in Bank documents:

- Always document the hazard scenarios, thresholds, damage curves, and cost assumptions used in your analysis so that results can be interpreted and reproduced.
- Label exported maps clearly with the hazard type, scenario, and threshold.
- Where results are presented to non-technical audiences, accompany bar charts with a brief narrative explaining what the exposure or damage percentages mean for the project.

Use Cases for World Bank Task Teams

The following examples illustrate how the tool can support different stages of the project cycle.

Pre-feasibility screening for infrastructure projects: A task team preparing a rural digital connectivity project can use the exposure analysis to screen proposed tower and fiber route locations against flood and cyclone hazard layers before commissioning detailed engineering studies. Early identification of high-risk sites allows the team to explore alternative routings or designs at low cost, before significant resources are committed.

Climate risk disclosure in Project Appraisal Documents: Under World Bank operational policies, task teams are required to assess and disclose climate risk for investment lending operations. The tool supports a rapid, documented screening that produces exportable maps and charts suitable for inclusion in PADs, Environmental and Social Assessments, or climate risk annexes. The structured workflow also helps teams demonstrate that a systematic assessment was conducted.

Investment prioritization for rehabilitation portfolios: For a digital infrastructure project with a portfolio of existing assets, the vulnerability-based damage assessment can help prioritize which assets to rehabilitate first, based on expected economic losses under current and future climate conditions, rather than relying solely on engineering condition surveys or ad hoc judgment.

Adaptive design for long-lived infrastructure: A task team designing a coastal or riverine infrastructure asset (such as a backbone fiber route) with a design life of 30–50 years can run the tool under multiple climate scenarios to understand how risk evolves over time. This can inform decisions about incorporating climate-adaptive design features, increasing engineering tolerances, or phasing investments.

Scenario comparison for climate-informed siting decisions: When a project involves choosing between two or more candidate sites, the tool can generate comparable exposure and damage estimates for each, providing a structured, evidence-based input into siting decisions that goes beyond engineering and cost criteria alone.

Support to post-disaster rapid assessments: Following a major weather event, the tool can be used to rapidly estimate which assets in an affected area were likely exposed above damage thresholds, based on pre-existing hazard data. This can help prioritize field assessment missions and support early loss estimation before teams are deployed.

Limitations and Important Caveats

Users should be aware of the following limitations when interpreting and communicating results.

- **Exposure analysis is binary.** The tool identifies assets as either above or below a threshold, it does not capture the degree of exposure or the spatial distribution of hazard intensity within an asset's footprint.
- **Damage estimates are indicative, not precise.** Results are sensitive to the choice of damage curve and replacement cost assumptions. They should be treated as relative risk indicators rather than engineering-grade loss estimates.
- **Hazard data quality varies by region.** Global hazard datasets perform better in some geographies than others. In data-sparse regions, results should be interpreted with particular caution and ideally validated against local knowledge or higher-resolution data.
- **The tool does not capture all dimensions of risk.** It does not assess cascading failures, service disruption costs, or social impacts. It should be used as one input among several in a broader climate risk assessment.
- **Damage curves are asset-type-specific.** Using a damage curve calibrated for a different asset type or region can produce misleading results. Document your sources clearly.

Getting Help and Providing Feedback

The tool is currently in beta. The development team welcomes your feedback on functionality, usability, and additional features that would support your work.

To request a demonstration, provide feedback, or ask a question, please contact [insert contact details].